# When Does Test-Time Adaptation Help in Smart Buildings? A Protocol-First Evaluation of Three Image-Domain Methods on Two Sensor Tasks

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ARTICLE INFO

*Keywords*:
Test-time adaptation
Human activity recognition
Occupancy estimation
Smart buildings
Evaluation protocol

ABSTRACT

Buildings consume a large share of global energy, and occupancy-aware control depends on models that keep working after they are installed somewhere new. Test-time adaptation (TTA) is an attractive answer: a trained model updates itself online from the unlabelled stream it is already processing, without source data and without annotation from the new site. The methods that define the area were developed for image corruptions, and whether their assumptions survive the move to one-dimensional sensor data is an open question.

We evaluate three of them — DIGA, PETAL and PeTTA — on two smart-building tasks: human activity recognition across two instrumented apartments, and occupancy estimation across two instrumented offices. Because the comparison is only as trustworthy as the protocol underneath it, we first quantify a measurement problem that is known in the activity recognition literature but still common in practice. Sliding windows with unit stride make adjacent samples share nineteen of twenty timesteps, so a random train/validation split places a near-duplicate of almost every validation sample into training. Under that split a source model reaches 0.998 macro-$F_1$; grouped by recording day it reaches 0.529. The window definition accounts for under two points of the difference and the split for forty-six. Every result we report uses day-grouped splits.

On that footing the cross-building penalty is asymmetric. Measured against a model trained on the target building at matched sample size, it is 0.6 macro-$F_1$ points entering one apartment and 21.3 entering the other. The three methods track the asymmetry rather than the task: in the direction with no measurable shift all three lose ground, by 8.6 to 20.4 points, and only DIGA gains anywhere, by 2.1 points in the direction with real headroom. A practitioner cannot know in advance which of the two situations a deployment is in, since establishing that would require the target labels TTA exists to do without.

The mechanism is shared. All three methods adapt batch-normalisation statistics, and that channel carries their behaviour: disabling it while leaving everything else intact reduces PETAL and PeTTA to altering fewer than 25 of 3412 predictions, and the cost of the channel rises monotonically with the share of target statistics mixed in, from −5.7 points at pure source statistics to −33.0 at pure target ones. We also find that the per-sensor features dominating the conventional design score *below* stratified-random guessing on the target in both directions, because a given sensor index is a different physical device in each building.

Occupancy estimation reproduces the pattern on a different task, dataset and sensing modality, and identifies the same cause independently. Two of the nine features shared between the offices correlate with occupancy in one and anti-correlate in the other; removing them moves cross-office transfer from MCC −0.38 to +0.43 and cuts the measured penalty in one direction from 47.2 points to 7.6. Against the checkpoint that those features break, PETAL and PeTTA gain over 50 MCC points; against every sound checkpoint they lose. We conclude that in this domain the effectiveness of test-time adaptation is governed by the quality of the checkpoint being adapted and by the reliability of the measurement, more than by the merits of the adaptation method.

## 1. Introduction

Buildings account for a large share of global energy demand, and occupancy-aware control of heating, ventilation and lighting is among the more direct levers available for reducing it [1, 2]. Human activity recognition (HAR) and occupancy estimation (OE) supply the signal such control needs [3, 4]. Both are learned from sensor traces, and both face the same obstacle when a model trained in one building is installed in another: room layouts differ, sensors sit in different places, and occupants keep different routines [5, 6]. A model that works where it was trained often does not work in the next building.

Test-time adaptation addresses this without requiring labels from the new site. A frozen source model updates itself online from the unlabelled stream it is already processing, with no access to source data and no target annotation [7]. For buildings, where sensors are always on and manual annotation is impractical, the fit is obvious. The method family has matured quickly on images: DIGA [8] adapts normalisation statistics and class prototypes without gradients, PETAL [9] treats lifelong adaptation as probabilistic inference, and PeTTA [10] stabilises adaptation when environments recur. Whether their assumptions survive the move

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to one-dimensional sensor data is an open question, and the one we set out to answer.

Answering it requires first establishing that the quantity these methods are asked to reduce is measured soundly, and we begin there. Features in this literature are computed over sliding windows with unit stride, so consecutive samples share nineteen of their twenty timesteps, and a random train/validation split places a 95%-identical copy of nearly every validation sample into training. That overlapping windows bias cross-validation in activity recognition is established [14]; what we add is the magnitude on this pipeline and the separation of the split from the window definition. We then evaluate the three methods on two tasks under the corrected protocol.

Our contributions are as follows.

- **A trustworthy evaluation protocol for this data type.** We isolate the measurement problem by varying the window definition and the validation split independently. Causality and segmentation move source macro-$F_1$ by under two points; the split moves it by forty-six (Section 4). Everything thereafter uses day-grouped splits, and the cross-building penalty is measured against a model trained on the target building at matched sample size rather than against source validation.
- **A rigorous evaluation of DIGA, PETAL and PeTTA across two tasks.** Both transfer directions, both tasks, published hyperparameters, chance baselines and paired bootstrap intervals throughout (Sections 5 and 6). The three methods track the measured penalty rather than the task: they help only where there is headroom, and lose ground where there is none.
- **A cross-task-confirmed account of why source models fail to transfer.** On activity recognition, features tied to a sensor index score below stratified-random guessing in both directions. On occupancy, two features reverse their correlation with the label between offices, and removing them changes the sign of the transfer result. The same failure — a feature whose meaning is specific to one installation — is found independently on two tasks, two datasets and two sensing modalities.

Section 2 reviews domain adaptation and TTA for sensor data. Section 3 describes the data and the three methods. Section 4 establishes the protocol. Sections 5 and 6 report the two evaluations. Sections 7 and 8 discuss and conclude.

## 2. Related work

### 2.1. Domain adaptation in smart buildings

Transferring occupancy and activity models between rooms has been studied extensively by Dridi and colleagues, through fine-tuning and layer freezing [3], unsupervised adaptation with and without source access [15], and clustering-based alignment [16]. Abderrahim et al. extend the setting to open and universal label spaces [17, 18]. These works establish that cross-room transfer is a real and recurring problem in this domain. They generally assume source data remains available, or that adaptation may run offline over the full target set, neither of which holds for a streaming deployment.

### 2.2. Test-time adaptation and its assumptions

Test-time adaptation drops both assumptions: the source model is frozen, source data is unavailable, and adaptation proceeds online over the target stream [7, 11, 12, 13]. The price is a set of implicit assumptions about what the shift looks like. Most of the family adapts normalisation statistics, which presumes the shift is largely a change in the first two moments of the features, and most presumes a test stream long enough for a slowly-updated teacher to move. Both presumptions are comfortable on image corruption benchmarks, where a corruption shifts an entire feature distribution coherently and a single corruption supplies tens of thousands of images. Neither is obviously safe when the shift is a change of building.

### 2.3. TTA for one-dimensional data

Work applying TTA to sensor streams is comparatively thin. Wang et al. report an optimisation-free method for cross-person activity recognition [19]. What is largely missing is an account of how evaluation protocol and feature design interact with measured adaptation gains, which is the gap this paper addresses. The closest prior evidence on the protocol side is Hammerla and Plötz [14], who show that pairwise similarity between overlapping windows biases cross-validation in activity recognition. That finding is a decade old and the practice it warns against remains common; Section 4 quantifies what it costs on the present pipeline.

## 3. Data, model and methods

### 3.1. Activity recognition

The dataset of Tapia et al. [20] provides two single-occupant apartments, denoted $S_1$ and $S_2$, instrumented with binary proximity sensors placed on household objects and furniture and sampled at 30 s. We use the five activities common to both: `toileting`, `meal_prep`, `watching_tv`, `music` and `cleaning`. After windowing there are 5031 samples from $S_1$ over 16 recording days and 3412 from $S_2$, with 53 sensor channels shared between the two apartments. Class sizes are uneven; `cleaning` has 442 windows in $S_1$ and 113 in $S_2$.

### 3.2. Occupancy estimation

The occupancy data comes from two offices at the Grenoble Institute of Technology, instrumented with a carbon dioxide sensor, a motion counter, an acoustic level (`rms`), door and window contacts, and several sub-metered power channels, sampled at 10 minutes. The source office contributes 1210 samples over 10 recording days and the target office 1041 over 8. Labels are occupant counts, which we binarise to occupied against unoccupied; the positive rate

is 33.5% and 28.0% respectively. Nine feature columns are common to both offices. This data is not ours to redistribute.

### 3.3. Model

All experiments use the same classifier: a three-layer perceptron with batch normalisation, ReLU and dropout at each layer, widths 256–128–64, and a linear output. The architecture is held fixed throughout so that differences between conditions are not confounded by capacity. Source checkpoints are selected by held-out macro-$F_1$ under the protocol of Section 4.

### 3.4. Adaptation methods

We evaluate three methods, each implemented from the equations and hyperparameters of its source paper and run at the configuration its released implementation uses.

**DIGA** [8] adapts without gradients. Its distribution adaptation module mixes frozen source batch-norm statistics with those of the current test batch under a weight $\lambda_{BN}$, and its semantic adaptation module maintains class prototypes from confident predictions and fuses a prototype-based posterior with the classifier's.

**PETAL** [9] treats lifelong adaptation as probabilistic inference. A student is updated against pseudo-labels from an exponential-moving-average teacher, with a term pulling the weights toward a SWAG posterior fitted on the source, and a Fisher-information criterion restoring parameters whose importance falls below a threshold.

**PeTTA** [10] targets recurring test environments. It maintains a category-balanced memory bank and senses divergence from the source feature distribution, using that signal to modulate both the teacher update rate and the strength of an anchor loss.

Two further methods appear throughout as *reference baselines* rather than as methods under evaluation. **AdaBN** replaces the source normalisation statistics with those of the test batch and does nothing else; **TENT** [11] adds entropy minimisation over the normalisation parameters. They are included because the three evaluated methods all contain a normalisation-adaptation component, and isolating what that component contributes requires a method that contains nothing else.

### 3.5. Metrics

Classes are unbalanced in both tasks, so we report macro-$F_1$ and Matthews correlation coefficient rather than accuracy alone, with accuracy and balanced accuracy beside them for the binary task. Two chance baselines accompany every activity result: stratified-random sampling from the target class distribution, and the majority class. Intervals on differences are paired bootstraps over the target stream.

## 4. Establishing a trustworthy protocol

Features in this literature are computed over a sliding window advanced one sample at a time. At the window length of $W = 20$ used here, consecutive feature vectors are built from overlapping spans sharing nineteen of twenty timesteps, so neighbouring samples are close to duplicates of one another. A validation split drawn at random over such a sequence therefore places a near-copy of almost every validation sample into the training set, and the resulting score measures memorisation rather than generalisation. Hammerla and Plötz [14] identified this effect in activity recognition a decade ago. Our purpose here is to quantify what it costs on the present pipeline and to separate it from a second choice it is easily confused with, namely how the window itself is defined.

**Table 1**
Source-domain macro-$F_1$ on $S_1$ under three validation splits, with the window definition held fixed. Target performance is unchanged across all three, so the difference is entirely in what the source figure measures.

| Validation split | Source macro-$F_1$ | Target macro-$F_1$ |
|---|---|---|
| Random over samples | $0.9984 \pm 0.001$ | 0.1626 |
| Blocked into runs | $0.4431 \pm 0.018$ | 0.1626 |
| Grouped by recording day | $\mathbf{0.5287} \pm 0.007$ | 0.1626 |

Two choices are varied independently. The first is causality: whether the window centred on a labelled sample extends symmetrically in both directions, which lets a feature read five minutes into the future and contradicts the online premise of test-time adaptation, or extends backwards only, which is what a deployed system can compute. The second is the validation split: random over samples, blocked into contiguous runs, or grouped by recording day so that no day contributes to both sides.

Table 1 and Figure 1 separate the two. Changing the window definition — causal against centred, and how segments are delimited — moves the source figure by under two points. Changing the split moves it by forty-six, from 0.9984 to 0.5287. The inflation is a property of the split, not of the window, and a pipeline that fixes the window while leaving the split random retains essentially all of it.

The practical consequence is that a cross-building drop computed as source validation minus target performance is not a measurement of two buildings. Under the random split that difference is 83 points on this data, almost all of which is the source figure being wrong. We therefore adopt grouped-by-day splits for every result that follows, and we make windows causal, which is required anyway if the features are to be computable online. The second change is not a trade: causal windows improve target macro-$F_1$ from 0.1626 to 0.2356, because a centred window straddling an activity boundary mixes two activities.

A further consequence shapes Section 5. If source validation cannot be trusted as a reference point, then the cross-building penalty should not be computed against it. We instead hold the evaluation set fixed and vary only where the training labels came from, comparing a model trained on the target building's own days with one trained on the source building, at matched training-set size. That quantity is the price of not having labels in the new building, which is what test-time adaptation is asked to recover.

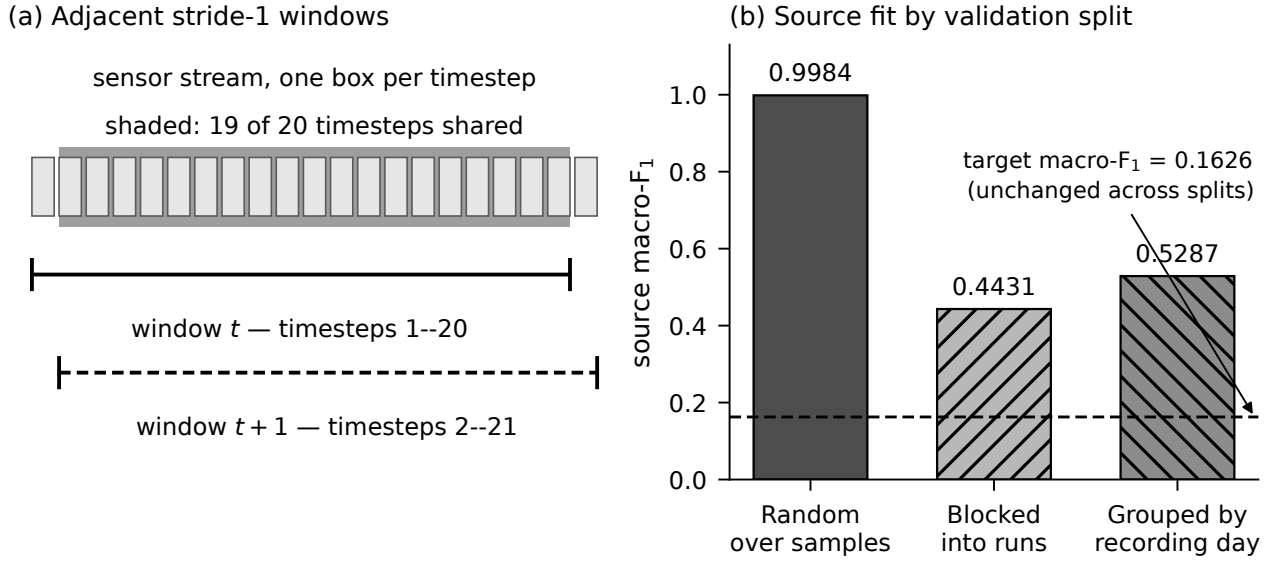



**Figure 1:** Overlapping-window leakage. Left: adjacent stride-1 windows share 19 of 20 timesteps. Right: source macro-$F_1$ under three validation splits; the window definition accounts for under two points of the difference, the split for forty-six.

**Table 2**
Target macro-$F_1$ by feature set, both transfer directions, causal features and day-grouped source selection. Stratified-random guessing scores 0.200 in both directions; figures below it are in bold.

| Feature set | $d$ | $S_1 \to S_2$ | $S_2 \to S_1$ |
|---|---|---|---|
| Layout-invariant | 13 | **0.4466** | **0.3068** |
| No per-sensor | 29 | 0.3630 | 0.2271 |
| Conventional (all) | 135 | 0.2118 | **0.1583** |
| Per-sensor only | 106 | **0.1350** | **0.1702** |
| Stratified random | — | 0.1997 | 0.2003 |

**Table 3**
Cross-building penalty on a fixed evaluation set at matched training-set size. *Own days* trains on the apartment being evaluated; *other building* trains on the other one. The difference is what a deployment pays for having no labels in the new building.

| Evaluated on | Own days | Other building | Penalty |
|---|---|---|---|
| $S_2$ held-out days | 0.3497 | 0.3434 | **0.6** pts |
| $S_1$ held-out days | 0.4897 | 0.2767 | 21.3 pts |

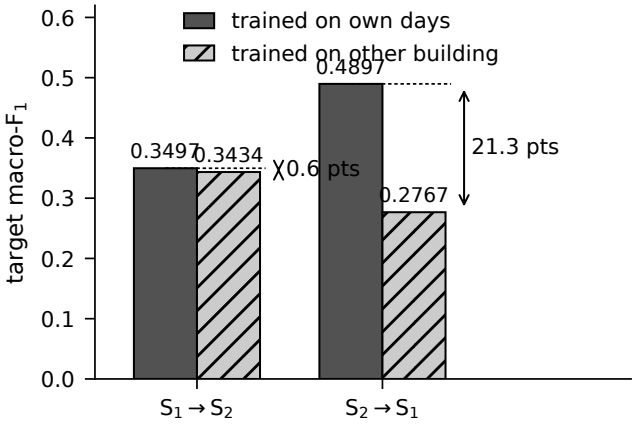



**Figure 2:** The cross-building penalty is asymmetric: negligible entering $S_2$, substantial entering $S_1$.

## 5. Activity recognition

### 5.1. Sensor identity does not transfer

The conventional feature vector for this data devotes most of its dimensions to per-sensor statistics: the mean and standard deviation of each sensor channel over the window. Such a feature encodes which device fired. Sensor index 100 is a different physical device in each apartment, so a rule learned about it is not merely uninformative in the other building but wrong.

Table 2 measures the consequence. Per-sensor features alone score 0.1350 and 0.1702 against a stratified-random baseline of approximately 0.200: below chance in both directions. They are not weakly informative; a model built on them is systematically wrong in the other apartment. The conventional 135-dimensional vector, which is dominated by them, falls below chance in one direction as well.

Replacing them with thirteen window-level summary statistics that describe how much activity is occurring and when, without naming any sensor, raises target macro-$F_1$ to 0.4466 and 0.3068. That is the checkpoint used for everything that follows. The result parallels the general principle that a feature tied to a specific installation cannot transfer, and Section 6.1 finds the same principle at work on a different task and modality.

### 5.2. The cross-building penalty is asymmetric

Table 3 and Figure 2 report the penalty measured as described in Section 4. It is not one number. Entering $S_2$ it is

**Table 4**
Target macro-$F_1$ for the three methods and the two reference baselines, both directions, published settings. Δ is against source-only in the same direction; *changed* counts altered predictions out of 3412. The lower block disables the normalisation-adaptation channel while leaving everything else unchanged.

| | $S_1 \to S_2$ (penalty 0.6) | | | $S_2 \to S_1$ (penalty 21.3) | | |
|---|---|---|---|---|---|---|
| Method | macro-$F_1$ | Δ | changed | macro-$F_1$ | Δ | changed |
| Source-only | 0.4944 | — | 0 | 0.3226 | — | 0 |
| DIGA | 0.4088 | −8.6 | 1279 | **0.3438** | **+2.1** | 1031 |
| PETAL | 0.3113 | −18.3 | 1739 | 0.2028 | −12.0 | 3105 |
| PeTTA | 0.2908 | −20.4 | 1845 | 0.2158 | −10.7 | 3164 |
| AdaBN *(baseline)* | 0.3139 | −18.0 | 1784 | 0.2137 | −10.9 | 3143 |
| TENT *(baseline)* | 0.3141 | −18.0 | 1782 | 0.2133 | −10.9 | 3142 |
| *normalisation channel disabled, everything else identical* | | | | | | |
| PETAL | 0.4947 | +0.0 | 11 | 0.3224 | −0.0 | 24 |
| PeTTA | 0.4947 | +0.0 | 5 | 0.3225 | −0.0 | 13 |

0.6 points, which against a between-fold standard deviation of 0.15 is not distinguishable from zero: a model that has never seen $S_2$ performs on $S_2$ as well as a model trained on $S_2$'s own days. Entering $S_1$ the same comparison costs 21.3 points.

This asymmetry organises the rest of this section, and it is worth being explicit about why both directions are reported rather than the more favourable one. In the direction with no measurable shift there is nothing for an adaptation method to recover, so any movement it produces can only be downward. A reader might reasonably ask why a practitioner would not simply deploy in the favourable direction. The answer is that the favourable direction is not identifiable in advance. Establishing which of the two situations a given deployment is in requires measuring the penalty, and measuring the penalty requires labels in the target building — which is precisely the thing test-time adaptation exists to avoid needing. A practitioner installing a model in a new building faces an unknown draw from this distribution of outcomes. Evaluating only the direction that happens to have headroom would therefore report a best case as though it were the expected one. Reporting both is not a limitation of the study; it is the result.

## 5.3. Method results and a shared mechanism

Table 4 gives the main activity result. In the direction where the penalty is 0.6 points, all three methods lose ground, by 8.6 to 20.4 macro-$F_1$ points. In the direction where it is 21.3, DIGA gains 2.1 points and the other two still lose. The methods track the measured penalty rather than the task: the only positive entry in the table is in the direction that has headroom.

The lower block identifies why. All three methods adapt batch-normalisation statistics, and in each released implementation that is the primary mechanism rather than an incidental detail. Disabling that channel and leaving every other component untouched — the gradient updates, the posterior regularisation, the divergence gating, the memory bank — reduces PETAL to altering 11 predictions of 3412 in one direction and 24 in the other, and PeTTA to 5 and 13. Whatever those components contribute, it is not visible at this scale. With the channel active, PETAL falls within 0.3 points of plain AdaBN and PeTTA within 2.3, having changed much the same predictions.

**Table 5**
DIGA as $\lambda_{BN}$ moves from source statistics (1.0) to target statistics (0.0), Δmacro-$F_1$ against source-only. At $\lambda_{BN} = 1.0$ the statistics are untouched, so that row isolates the prototype classifier from the normalisation.

| $\lambda_{BN}$ | $S_1 \to S_2$ | $S_2 \to S_1$ |
|---|---|---|
| 1.0 (source) | −5.7 | **+4.4** |
| 0.9 | −6.5 | +2.4 |
| 0.8 (published) | −8.6 | +2.1 |
| 0.6 | −23.3 | −12.9 |
| 0.4 | −30.3 | −19.9 |
| 0.2 | −25.9 | −22.8 |
| 0.0 (target) | −33.0 | −20.5 |

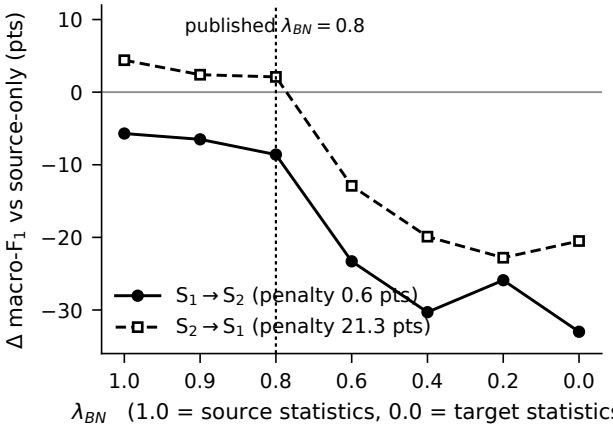



**Figure 3:** Cost of the shared normalisation-adaptation mechanism. Moving from source statistics ($\lambda_{BN} = 1.0$) to target statistics ($\lambda_{BN} = 0.0$) degrades performance monotonically in both directions.

**Table 6**
Correlation with occupancy per office. Two of the nine shared features have 95% bootstrap intervals of opposite sign that do not overlap, over 2000 resamples: `rms` is $[+0.595, +0.666]$ against $[-0.316, -0.111]$, and `derivative_co2` is $[+0.057, +0.172]$ against $[-0.295, -0.123]$.

| Feature | Source office | Target office |
|---|---|---|
| `sum_power` | +0.502 | +0.780 |
| `co2` | +0.378 | +0.671 |
| `power1` | +0.352 | +0.462 |
| `motion` | +0.348 | +0.544 |
| `power2` | +0.263 | +0.423 |
| `door` | +0.255 | +0.307 |
| `power3` | +0.130 | +0.721 |
| `rms` | **+0.626** | **−0.202** |
| `derivative_co2` | +0.116 | −0.207 |

The channel is a continuum rather than a switch, and DIGA's $\lambda_{BN}$ exposes it directly. Table 5 and Figure 3 sweep it. The cost rises monotonically with the share of target statistics mixed in, in both directions, and steeply: moving from the published 0.8 to 0.6 costs about fifteen points either way. The top row is also informative. At $\lambda_{BN} = 1.0$ the statistics are exactly the source model's, yet 1026 predictions change, because DIGA's prototype classifier is still active. That row is the only configuration in this study where an adaptation component helps by more than a point without target labels, and it helps in the direction with headroom and hurts in the direction without. DIGA's small positive result is therefore a prototype classifier gaining 4.4 points, partly spent back by a normalisation step costing 2.3.

One objection deserves pre-empting: that the methods were simply not tuned. We searched each method's own published hyperparameter grid in both directions — 64 configurations for DIGA, 24 for PETAL, 16 for PeTTA. The conclusion does not move. In the direction without headroom 53 of DIGA's 64 settings lose ground and the best gains 1.1 points; no setting anywhere in PETAL's or PeTTA's published space alters more than 38 of 3412 predictions, so their behaviour is a property of the update rules rather than of the configuration.

## 6. Occupancy estimation

This section asks whether the activity result is a property of that dataset or of the problem. The task changes from five-class activity recognition to binary occupancy, the sensing changes from binary proximity switches to carbon dioxide, acoustics, motion and sub-metered power, the environments change from apartments to offices, and the data comes from a different institution. If the pattern of Section 5 is about cross-building transfer rather than about one dataset, it should reappear.

**Table 7**
Cross-office penalty in MCC, day-grouped folds and matched training-set size, measured exactly as in Table 3.

| Features | Evaluated on | Own days | Other office | Penalty |
|---|---|---|---|---|
| All 9 | target | 0.7867 | 0.2138 | 57.3 |
| All 9 | source | 0.7643 | 0.2924 | 47.2 |
| 7 transf. | target | 0.8225 | 0.3777 | 44.5 |
| 7 transf. | source | 0.3624 | 0.2866 | **7.6** |

### 6.1. The penalty, and a feature that means two different things

Table 6 reports how each shared feature relates to occupancy in each office. Seven behave consistently. Two do not. The acoustic level `rms` is the strongest single predictor the source office has, at +0.626, and in the target office it points the other way at −0.202, with non-overlapping intervals. The rate of change of carbon dioxide does the same more weakly.

It is worth being precise about what kind of defect this is, because the obvious description is wrong. These are not noisy features to be pruned. Removing both drops the source office's own-days MCC from 0.7643 to 0.3624, so nearly half of what that office knows about its own occupancy is carried by a feature that means the opposite thing next door. The feature is genuinely predictive at home and genuinely misleading away. That combination is exactly what makes it invisible to any evaluation conducted in a single building, and it is why the defect survives into a cross-building result rather than being caught upstream.

This is the same failure as Section 5.1, reached independently. There the quantity that did not transfer was a sensor index; here it is a sensor placement. In both cases a feature encodes something specific to one installation, and in both cases the consequence is not a weaker model elsewhere but a wrong one.

Table 7 applies the measurement of Section 4 to this task; the raw sensor records carry timestamps, giving ten recording days in the source office and eight in the target. The penalty is again asymmetric, and dropping the two reversing columns cuts it into the source office from 47.2 points to 7.6. Most of what presented as a cross-office domain gap in that direction was two features meaning opposite things in the two rooms. Into the target office a penalty of 44.5 points survives the same surgery, so unlike the apartment pair these two environments do contain a real shift in one direction.

### 6.2. The methods reproduce the pattern

Table 8 runs the three methods and the two baselines on this task, in a single online pass over the target office in recording order, each batch predicted before the model updates on it, with the scaler fitted on the source split alone and no target labels used for anything.

The pattern of Section 5.3 reproduces. Against the checkpoint that the reversing features break — MCC −0.3818, meaning its predictions are anti-correlated with the labels — PETAL gains 51.9 MCC points and PeTTA 50.9. Against the sound checkpoint in the other direction they lose 15.9

**Table 8**
Occupancy, MCC, single online pass in recording order with the scaler fitted on the source split and no target labels used anywhere. Δ is against source-only in the same cell.

| Features | Direction | Source-only | DIGA | PETAL | PeTTA | AdaBN *(b)* | TENT *(b)* |
|---|---|---|---|---|---|---|---|
| 9 shared | source → target | −0.3818 | +0.075 | **+0.519** | **+0.509** | +0.521 | +0.521 |
| 9 shared | target → source | +0.3579 | +0.011 | −0.159 | −0.168 | −0.119 | −0.119 |
| 7 transferable | source → target | +0.4314 | +0.110 | +0.122 | +0.028 | +0.083 | +0.083 |
| 7 transferable | target → source | +0.3465 | +0.019 | −0.009 | +0.005 | +0.046 | +0.046 |

and 16.8. The gains are large, and they are large because the starting point is broken.

The lower two rows make that explicit in a way the activity task could not. Restricted to the seven transferable features, no checkpoint is anti-correlated and no method gains more than 12.2 points. The adaptation gain and the feature defect appear and disappear together. On the activity task we could show that adaptation gain correlates with checkpoint quality; here the checkpoint's deficiency has a named cause, and removing the cause removes the gain.

### 6.3. Metrics that reward the wrong thing

A binary task at roughly 30% positives rewards predicting the majority class, and the standard check is whether a method has collapsed to it. None here has: predicted-positive rates stay between 17% and 62% in all twenty-four cells. Two subtler effects do appear, and both are worth reporting because a narrower set of metrics would hide them.

First, four of the five adaptation conditions in the target → source direction improve recall on the occupied class while losing MCC: AdaBN takes occupancy recall from 0.355 to 0.428 while MCC falls from +0.358 to +0.239, and PETAL and PeTTA do the same. Occupancy detection rate is a natural headline for this task, and a report using it alone would score those conditions as improvements.

Second, the source-only model in the seven-feature source → target cell attains occupancy recall 0.967 and predicts occupied in 62% of windows against a true rate of 28%. It has over-committed to the *minority* class. Its accuracy of 0.640 sits below the 0.720 that predicting "unoccupied" everywhere would earn, while its MCC of 0.431 looks reasonable. A collapse check that looks only for majority-class degeneracy passes this model without comment. Both classes' recalls are needed to see what it is doing, which is why they are reported throughout.

## 7. Discussion

Two things govern whether test-time adaptation helps on this data, and neither is a property of the adaptation method.

The first is the quality of the checkpoint being adapted. Across both tasks and both directions, the methods gain where the starting model is poor and lose where it is sound. The clearest instance is occupancy, where a checkpoint whose predictions are anti-correlated with the labels yields gains above 50 MCC points for two of the three methods, while the sound checkpoint in the other direction yields losses of 16. The activity task shows the same ordering more gently. The practical reading is that an adaptation gain reported against a single checkpoint is, in large part, a statement about that checkpoint. A method evaluated only against a weak source model will appear to work, and the appearance will not survive a better one.

The second is the reliability of the measurement. A cross-building drop computed as source validation minus target performance was 83 points on this pipeline; measured against a model trained on the target building at matched sample size it is 0.6 points in one direction and 21.3 in the other. The quantity these methods are asked to reduce is therefore not well defined until the evaluation protocol is settled, and the two are easily conflated. A great deal of what looked like domain shift was a validation split, and on the occupancy task a further large share of it was two features whose meaning differs between rooms.

The mechanism by which all three methods act is the same, and it is narrower than the methods' descriptions suggest. Each adapts batch-normalisation statistics, and that channel carries their behaviour: with it disabled, PETAL and PeTTA alter fewer than 25 of 3412 predictions, and with it enabled they land within a few points of a baseline that does nothing else. Replacing source normalisation statistics with target ones is a trade, not an improvement: it discards a calibration fitted on labelled data in favour of one estimated from an unlabelled batch, and whether the trade pays depends on which of the two is closer to correct. On corruption benchmarks the target statistics are usually better, because a corruption shifts the feature distribution coherently. A change of building is not that kind of shift. The one component in this study that helped without target labels — DIGA's prototype classifier, worth 4.4 points in the direction with headroom — is the one that does not touch normalisation statistics at all.

*Recommendations.* Four follow directly.

Group validation splits by recording session, not at random, whenever features are computed over overlapping windows. The cost of not doing so was forty-six points here and is invisible in the reported figure.

Measure the cross-domain penalty against a model trained on the target domain at matched sample size, rather than against source validation. The two differ by a factor that depends on how hard each domain is, and only the former is a statement about transfer.

Prefer features that describe the environment's behaviour over features that name its instruments. A sensor index or an acoustic channel that is informative in one installation can be anti-informative in the next, and a chance baseline is what makes that visible.

Report an adaptation gain beside the quality of the checkpoint it was measured against, and in both transfer directions where two are available. A single favourable direction is a best case presented as an expectation.

## 8. Conclusion

We set out to evaluate three image-domain test-time adaptation methods — DIGA, PETAL and PeTTA — on human activity recognition and occupancy estimation across buildings, and this paper reports three results.

The first is a protocol. Sliding windows with unit stride make a random validation split place near-duplicates across it, inflating source macro-$F_1$ from 0.529 to 0.998 on this pipeline; the window definition accounts for under two points of that and the split for forty-six. Grouping the split by recording day removes the effect, and measuring the cross-building penalty against a target-trained model at matched sample size gives a quantity that means what it appears to mean. On that footing the penalty is 0.6 macro-$F_1$ points entering one apartment and 21.3 entering the other.

The second is the evaluation. The three methods track that asymmetry rather than the task. Where the penalty is negligible all three lose ground, by 8.6 to 20.4 points; where it is real, only DIGA gains, by 2.1. Occupancy estimation reproduces the pattern on a different task, dataset and sensing modality: against a checkpoint whose predictions are anti-correlated with the labels, PETAL and PeTTA gain over 50 MCC points, and against a sound one they lose 16. The shared mechanism is normalisation-statistic adaptation, which carries essentially all of their behaviour and which is a trade rather than an improvement.

The third is an account of why source models fail to transfer, confirmed independently on both tasks. On activity recognition, features that name a sensor score below stratified-random guessing in both directions, because an index identifies a different device in each apartment. On occupancy, two of nine shared features reverse their correlation with the label between offices; removing them moves cross-office transfer from MCC $-0.38$ to $+0.43$ and cuts the measured penalty in one direction from 47.2 points to 7.6. The same failure appears in both: a feature whose meaning is specific to one installation is not weakly informative elsewhere but actively wrong, and nothing in a single-building evaluation can reveal it.

We do not conclude that test-time adaptation is unsuitable for smart buildings. We conclude that in this domain its effectiveness is governed by the quality of the checkpoint being adapted and by the reliability of the measurement, more than by the merits of the adaptation method, and that a reported gain should be read in that light.

*Limitations.* The study rests on two apartments from one activity dataset and two offices from one occupancy dataset. Running both transfer directions on each rules out a direction-specific fluke but says nothing about other buildings, other sensor modalities or richer activity sets. The specific penalties we report belong to these pairs. What we would expect to transfer is the procedure: group the split by session, measure the penalty against a target-trained model at matched sample size, compute a chance baseline for every feature set, and report both directions.

## Reproducibility

Code and every result file are available at `https://github.com/mariemkbaier/TTA-Test-Time-Adaptation`. Each table above names the script and result file that produced it. The occupancy data was collected at the Grenoble Institute of Technology and is not ours to redistribute, so that section is reported but not independently reproducible from the released code.

## Acknowledgements

This research was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC).

## References

[1] M. Manfren, M. Sibilla, L. Tronchin, Energy modelling and analytics in the built environment—a review of their role for energy transitions in the construction sector, Energies 14 (3) (2021) 679.

[2] B. Amangeldy, T. Imankulov, N. Tasmurzayev, G. Dikhanbayeva, Y. Nurakhov, A review of artificial intelligence and deep learning approaches for resource management in smart buildings, Buildings 15 (15) (2025) 2631.

[3] J. Dridi, M. Amayri, N. Bouguila, Transfer learning for estimating occupancy and recognizing activities in smart buildings, Building and Environment 217 (2022) 109057.

[4] M. Amayri, S. Ploix, N. Bouguila, F. Wurtz, Estimating occupancy using interactive learning with a sensor environment: real-time experiments, IEEE Access 7 (2019) 53932–53944.

[5] T. Hong, D. Yan, S. D'Oca, C. Chen, Ten questions concerning occupant behavior in buildings: the big picture, Building and Environment 114 (2017) 518–530.

[6] R. Shahbazian, I. Trubitsyna, Human sensing by using radio frequency signals: a survey on occupancy and activity detection, IEEE Access 11 (2023) 40878–40904.

[7] J. Liang, R. He, T. Tan, A comprehensive survey on test-time adaptation under distribution shifts, International Journal of Computer Vision 133 (1) (2025) 31–64.

[8] W. Wang, Z. Zhong, W. Wang, X. Chen, C. Ling, B. Wang, N. Sebe, Dynamically instance-guided adaptation: a backward-free approach for test-time domain adaptive semantic segmentation, in: CVPR, 2023, pp. 24090–24099.

[9] D. Brahma, P. Rai, A probabilistic framework for lifelong test-time adaptation, in: CVPR, 2023, pp. 3582–3591.

[10] T.-H. Hoang, D. M. Vo, M. N. Do, Persistent test-time adaptation in recurring testing scenarios, in: NeurIPS, Vol. 37, 2024.

[11] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell, Tent: fully test-time adaptation by entropy minimization, in: ICLR, 2021.

[12] Q. Wang, O. Fink, L. Van Gool, D. Dai, Continual test-time domain adaptation, in: CVPR, 2022, pp. 7201–7211.

[13] J. Liang, D. Hu, J. Feng, Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation, in: ICML, Vol. 119 of PMLR, 2020, pp. 6028–6039.

[14] N. Y. Hammerla, T. Plötz, Let's (not) stick together: pairwise similarity biases cross-validation in activity recognition, in: Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing, 2015, pp. 1041–1051.

[15] J. Dridi, M. Amayri, N. Bouguila, Unsupervised domain adaptation without source data for estimating occupancy and recognizing activities in smart buildings, Energy and Buildings 303 (2024) 113473.

[16] J. Dridi, M. Amayri, N. Bouguila, Unsupervised clustering-based domain adaptation for estimating occupancy and recognizing activities in smart buildings, Journal of Building Engineering 85 (2024) 108741.

[17] O. Abderrahim, J. Dridi, M. Amayri, N. Bouguila, Occupancy estimation and activity recognition in smart buildings using open set domain adaptation, Journal of Building Engineering 112 (2025) 113784.

[18] O. Abderrahim, J. Dridi, M. Amayri, N. Bouguila, Scalable activity recognition in smart buildings via generalized domain adaptation of IoT sensor data, Energy and Buildings 351 (2026) 116692.

[19] S. Wang, J. Wang, H. Xi, B. Zhang, L. Zhang, H. Wei, Optimization-free test-time adaptation for cross-person activity recognition, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 7 (4) (2023) 183.

[20] E. M. Tapia, S. S. Intille, K. Larson, Activity recognition in the home using simple and ubiquitous sensors, in: Pervasive Computing, Springer, Berlin, Heidelberg, 2004, pp. 158–175.